Sec: OSR.IIT_*COSC(MODEL-A&APEX) | GTA-8(P2) | Date: 23-04-26
Time: 3 Hrs | 2025_P2 | Max. Marks: 180

KEY SHEET

KEY SHEET

MATHEMATICS

1	**B**	2	**D**	3	**C**	4	**B**	5	**AC**
6	**AB**	7	**ABC**	8	**ACD**	9	**95**	10	**1**
11	**3**	12	**9**	13	**490**	14	**5**	15	**0.06**
16	**15.08**								

PHYSICS

17	**A**	18	**D**	19	**C**	20	**C**	21	**ABC**
22	**BCD**	23	**BCD**	24	**AD**	25	**12.95 TO 12.98**	26	**0.16 TO 0.17**
27	**21.40 TO 21.45**	28	**1.20**	29	**3.45 TO 3.50**	30	**20**	31	**0.83**
32	**1042.38 TO 1042.42**								

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CHEMISTRY

33	**D**	34	**A**	35	**A**	36	**B**	37	**ABD**
38	**ACD**	39	**ABC**	40	**D**	41	**357**	42	**13**
43	**3**	44	**2**	45	**8**	46	**3**	47	**40.40**
48	**6650**								

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SOLUTIONS
MATHEMATICS

1. $\lim_{x\to 0^+}\left(\frac{e^{x\ln(2^x-1)}-(2^x-1)^x\sin x}{x^x}\right)^{\frac{1}{x}}=\lim_{x\to 0^+}\left(\frac{(2^x-1)^x(1-\sin x)}{x^x}\right)^{\frac{1}{x}}$

$=\lim_{x\to 0^+}\left(\frac{2^x-1}{x}\right)(1-\sin x)^{\frac{1}{x}}=(\ln 2)\lim_{x\to 0}e^{\frac{-\sin x}{x}}=(\ln 2)(e^{-1})$

2. $g'(\theta)=\left(f(\sin^2\theta)-f(\cos^2\theta)\right)\sin 2\theta\ x\in\left(\frac{-\pi}{2},\frac{-\pi}{4}\right)\Rightarrow\tan^2\theta>1\Rightarrow\sin^2\theta>\cos^2\theta$

$\Rightarrow f(\sin^2\theta)>f(\cos^2\theta)$ Also $\sin 2\theta<0\Rightarrow g'(\theta)<0$

$x\in\left(\frac{-\pi}{4},0\right)\Rightarrow g'(\theta)=(-)(-)=+\Rightarrow g(\theta)$ is increasing

$x\in\left(0,\frac{\pi}{4}\right)\Rightarrow g'(\theta)=(-)(+)=-\ x\in\left(\frac{\pi}{4},\frac{\pi}{2}\right)\Rightarrow g'(\theta)=(+)(+)=+$

3. If $n(P\cap Q)=0,n(PUQ)=4$ then P(no success $)=0$

If $n(P\cap Q)=1,n(PUQ)=3$ then P (no success)

$=\frac{{}^4C_1\,{}^3C_2\times 2}{({}^4C_2)^2}\times\frac{1}{4}=\frac{1}{6}$ If $n(P\cap Q)=2=n(P\cup Q)$ then P(no success)

$=\frac{{}^4C_2}{({}^4C_2)^2}\times\frac{2}{4}=\frac{1}{12}$ Required prob $=\frac{1}{6}+\frac{1}{12}=\frac{1}{4}$

4. $|\hat{u}+\hat{v}+2\hat{u}\times\hat{v}|^2=4\Rightarrow 1+1+4\sin^2\theta+2\cos\theta=4$

$\Rightarrow 4\cos^2\theta-2\cos\theta-2=0\Rightarrow\cos\theta=-\frac{1}{2},1\Rightarrow\cos\theta=-\frac{1}{2}\Rightarrow\theta=120^\circ$

$\Rightarrow\frac{|\hat{u}-\hat{v}|}{|\hat{u}\times\hat{v}|}=\frac{2\sin\frac{\theta}{2}}{\sin\theta}=\frac{2\sin 60^\circ}{\sin 120^\circ}=2$

5. $q+ip=e^{xee^{i\alpha}}\Rightarrow\frac{dq}{dx}+i\frac{dp}{dx}=e^{xee^{i\alpha}}\cdot e^{i\alpha}$

$\Rightarrow\left(\frac{dq}{dx}\right)^2+\left(\frac{dp}{dx}\right)^2=\left|e^{xei\alpha}\right|\left|e^{i\alpha}\right|=\left|e^{x(\cos\alpha+i\sin\alpha)}\right|\times 1=e^{x\cos\alpha}=p^2+q^2$ Also

$f(\alpha)=\int_0^1(p^2+q^2)dx=\int_0^1 e^{2x\cos\alpha}dx=\frac{1}{2\cos\alpha}\left|e^{2x\cos\alpha}\right|_0^1=\frac{e^{2\cos\alpha}-1}{2\cos\alpha}\Rightarrow f(0)=\frac{e^2-1}{2}$

6. $I=\frac{1}{5}\int\frac{dx}{\sin x-1}-\frac{1}{5}\int\frac{dx}{\sin x+4}$

$=\frac{1}{5}\int\frac{\sec^2\frac{x}{2}dx}{2\tan\frac{x}{2}-1-\tan^2\frac{x}{2}}-\frac{1}{5}\int\frac{\sec^2\frac{x}{2}dx}{2\tan\frac{x}{2}+4\tan^2\frac{x}{2}}=\frac{2}{5}\frac{1}{\left(\tan\frac{x}{2}-1\right)}-\frac{2}{5\sqrt{5}}\tan^{-1}\left(\frac{4\tan\frac{x}{2}+1}{\sqrt{15}}\right)+c$

7. $x^4-nx+63=(x-\alpha)(x^3+ax^2+bx+c)$

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Product $=63 \Rightarrow \alpha$ can be $1,3,7,9,63$

$a-\alpha=0,\ b-a\alpha=0, c-b\alpha=-n, -c\alpha=63 \Rightarrow n=\alpha^3+\frac{63}{\alpha}\ \alpha=1 \Rightarrow n=64\ \alpha=3 \Rightarrow n=48$ least value is $48=2^4 3^1$

$\Rightarrow$ no. of divisions = 10

$ab=48$ has solutions $\left({}^{4+2-1}C_4\right)\left({}^{1+2-1}C_1\right)=10$

8. $f(n)={}^{n}C_0\left({}^{n+1}C_1+{}^{n+1}C_2+\ldots+{}^{n+1}C_{n+1}\right)+{}^{n}C_1\left({}^{n+1}C_2+{}^{n+1}C_3+\ldots+{}^{n+1}C_{n+1}\right)+\ldots\ldots\ldots$

$+{}^{n}C_n\left({}^{n+1}C_{n+1}\right)=\left({}^{n}C_0\,{}^{n+1}C_1+{}^{n}C_1\,{}^{n+1}C_2+\ldots+{}^{n}C_n\,{}^{n+1}C_{n+1}\right)+\left({}^{n}C_0\,{}^{n+1}C_2+{}^{n}C_1\,{}^{n+1}C_3+\ldots+{}^{n}C_{n-1}\,{}^{n+1}C_{n+1}\right)$

$+\ldots\ldots$

$={}^{2n+1}C_n+{}^{2n+1}C_{n-1}+\ldots.+{}^{2n+1}C_0=2^{2n}$

9. Tangent is $y-k=-2(h+1)(x-h)$

$\Rightarrow A\left(h+\frac{k}{2(h+1)},0\right)$ at $P(h,k)$

and $B\left(0,k+2h(h+1)\right)$

$S(h)=\frac{1}{2}\times OA\times OB=\frac{1}{2}\left(\frac{k+2h^2+2h}{2(h+1)}\right)\left(k+2h^2+2h\right)=\frac{\left(h^2+4\right)^2}{4(h+1)}\ S'(h)=0\Rightarrow 3h^2+4h-4=0$

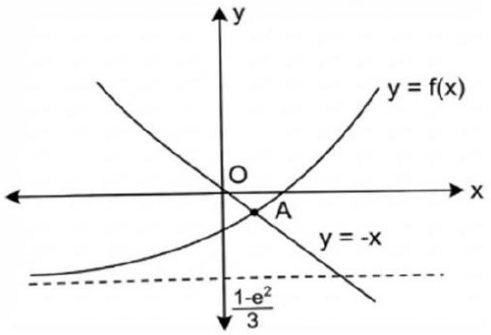
$\Rightarrow h=-2,\frac{2}{3}$ As $h\in(0,1)\Rightarrow h=\frac{2}{3}$

$\Rightarrow$ minimum value of $S(h)=S\left(\frac{2}{3}\right)=\frac{80}{27}$

$\Rightarrow \frac{10q-p}{2}=95$

10. Let $\int_0^2 f(x)\,dx=k$

$\Rightarrow \frac{dy}{dx}=y+k\Rightarrow\int\frac{dy}{y+k}=\int dx\Rightarrow \ln(y+k)=x+C_1\Rightarrow y+k=ce^x$



$\int_0^2 f(x)\,dx=k\Rightarrow k=\frac{C\left(e^2-1\right)}{3}$

$f(0)=\frac{4-e^2}{3}\Rightarrow C=1, k=\frac{e^2-1}{3}$

$\Rightarrow f(x)=e^x-\left(\frac{e^2-1}{3}\right)$

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Point A is only
solution of $f(x)=-x$

11. $2a^2b^3c^4=ab^2c^3+a^3b^4c^5$
$\Rightarrow 2ab^2c^3=bc^2+a^2\ b^3c^4$
$\Rightarrow 2abc=1+a^2\ b^2c^2$
$\Rightarrow (abc-1)^2=0$
$\Rightarrow abc=1$
$AM\geq GM\Rightarrow \frac{a+b+c}{3}\geq (abc)^{\frac{1}{3}}$
$\Rightarrow a+b+c\geq 3$

12. $y=3x+4$ cuts parabola at $A(-1,1)\ \&\ B(3,13)$
$y=mx+1$ cuts parabola at $C(0,1)\ \&$
$D(m-1,m^2-m+1)$
circle through A,B,C is $x^2+y^2+x-15y+14=0$
$D(m-1,m^2-m+1)$ lies on circle
$\Rightarrow m^4-2\ m^3-11\ m^2+12\ m=0$
$\Rightarrow m=0,1,4,-3$
$m=0\Rightarrow D(-1,1)$
$m=1\Rightarrow D(0,1)$
$m=3\Rightarrow D(3,13)$
rejected as points are distinct
$m=-3\Rightarrow D(-4,13)\Rightarrow$ correct

13. $0+1+2+3=6$
so remaining two digits has sum 4
$4=1+3,2+2,4+0$
$\Rightarrow \left(\frac{5!}{2!}+\frac{5!}{2!2!}+\frac{5!}{2!}\right)+\left(\frac{5!}{3!}+\frac{5!}{2!}+\frac{5!}{3!}\right)$
$+4\left(\frac{5!}{2!}\right)$
$=150+100+240=490$

14. $x^2+ay^2+4xy+(2b+1)x+30y+50$
$=(y-mx-C_1)(y-mx-C_2)$
$\Rightarrow -2\ m=\frac{4}{a},m^2=\frac{1}{a},-m(C_1+C_2)=\frac{2\ b+1}{a}$
$\frac{30}{a}=C_1+C_2,\frac{50}{a}=C_1C_2$
$\Rightarrow a=4,\ m=\frac{1}{2},C_1=5,C_2=\frac{5}{2},\ b=7$
$\text{distance}=\left|\frac{C_1-C_2}{\sqrt{1+m^2}}\right|=\left|\frac{5-\frac{5}{2}}{\sqrt{1+\frac{1}{4}}}\right|=\sqrt{5}$

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15. $|A^T||A||B| = |A||B||B^T|$

$\Rightarrow |A|^3 |A^TB| = |B||AB^T|$

$\Rightarrow |A|^3 = |B|^3 \Rightarrow |A| = 2$

$$\left|AB^{-1}\text{adj}\left(A^TB\right)^{-1}\right| = \frac{\left|AB^{-1}\right|}{\left|\text{adj}\left(A^TB\right)\right|}$$

$$= \frac{\left|AB^{-1}\right|}{\left|A^TB\right|^2} = \frac{|A|}{|B|} \times \frac{1}{|A^T|^2 |B|^2} = \frac{1}{|A||B|^3} = \frac{1}{16}$$

16. $x_1 = \sqrt{5} \Rightarrow V(5,0)$

$\Rightarrow$ Parabola is $y^2 = -4a(x-5)$

solve with ellipse $\Rightarrow \frac{x^2}{9} - 1 = a(x-5)$

$\Rightarrow x^2 - 9ax + 45a - 9 = 0$

$D = 0 \Rightarrow a = 2, \frac{2}{9}$

$a = 2 \Rightarrow x = -9$ rejected

$a = \frac{2}{9} \Rightarrow x = 1$

$\Rightarrow A\left(1, \frac{4\sqrt{2}}{3}\right), B\left(1, \frac{-4\sqrt{2}}{3}\right)$

tangent at A is $3x + 9\sqrt{2}y = 27 \Rightarrow C(9,0)$

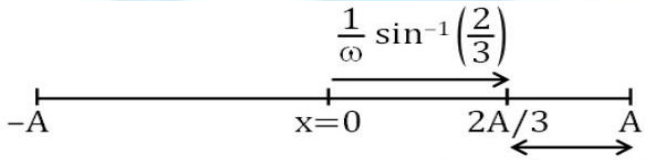
area of $\triangle ABC = \frac{1}{2} \times 8 \times \frac{8\sqrt{2}}{3}$

$= \frac{32\sqrt{2}}{3} = 15.0849$

PHYSICS

17. $x = A\sin(\omega t)$

$x \geq \frac{2A}{3}$



$$\frac{\pi}{2\omega} - \frac{1}{\omega}\sin^{-1}\left(\frac{2}{3}\right)$$

$$\text{Probability} = \frac{\left(\frac{\pi}{2} - \sin^{-1}\left(\frac{2}{3}\right)\right)\frac{1}{\omega}}{\frac{\pi}{\omega}} = \frac{1}{2} - \frac{1}{\pi}\sin^{-1}\left(\frac{2}{3}\right)$$

18. Length of mirror required doesn't depend upon distance of mirror from person.

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19. Energy required to produce one Au nucleus

$e=\left(M_{Au}+M_{Pb}+3m_P+8m_n\right)\times c^2$

$=(196.96657+3\times1.007276+8\times1.008665-$

$207.975000)\times931.5\text{MeV}$

$\approx 77.05\text{MeV}$

Energy required to produce 1g gold

$$E=\frac{1}{197}\times6\times10^{23}\times77.05\times10^{6}\times1.6\times10^{-19}\ \text{J}$$

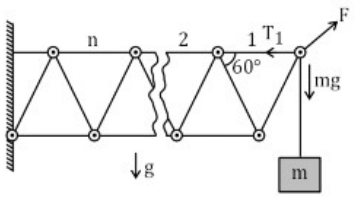
$$=3.75\times10^{10}\ \text{J}$$

$$=\frac{3.75\times10^{10}}{1000\times3600}\ \text{kW}-\text{hr}=\frac{3.75}{3.6}\times10^{4}\ \text{kW}-\text{hr}$$

$$=1.04\times10^{4}\ \text{kW}-\text{hr}$$

$$\Rightarrow \quad \text{cost}=1.5\times1.04\times10^{4}\text{Rs}=15625\text{Rs}$$

20.

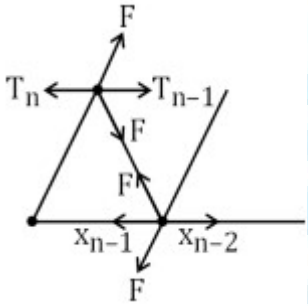


$T_1=\frac{F}{2}$

$mg=\sqrt{3}\frac{F}{2}$

$F=\frac{2mg}{\sqrt{3}}$

$T_1=\frac{mg}{\sqrt{3}}$



$T_n=T_{n-1}+F$

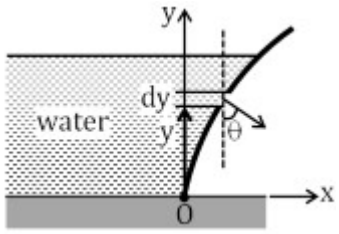
All slanted line have same magnitude of tension / compression.

$x_{n-2}=x_{n-1}+F$

$T_n=T_1+(n-1)\frac{2mg}{\sqrt{3}}=\frac{mg}{\sqrt{3}}(2n-1)$

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21.



$$F_H = \rho_w g \times \frac{h}{2} \times h \times b$$

$$= 1000 \times 10 \times \frac{2.5}{2} \times 2.5 \times 8 \text{ N}$$

$$= 250\text{KN}$$

$$y^2 = 5x \Rightarrow 2yy' = 5$$

$$\tan\theta = y' = \frac{5}{2y}$$

$$F_v = \int \rho_g (h - y) \times W \times dy \cot\theta$$

$$= 10^4 \times 8 \int_0^h \frac{2y}{5}(h - y)\,dy$$

$$= 32\text{KN} \int_0^h (hy - y)\,dy$$

$$= 32 \times \frac{h^3}{6} = 16 \times \frac{2.5 \times 2.5 \times 2.5}{6}\text{KN}$$

$$= \frac{250}{3}\text{KN}$$

$$\tau = \int \rho g y (h - y) W (y dy + x \cot\theta dy)$$

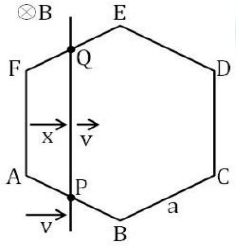
$$= 10^4 \times 8 \int_0^h \left[y^2 (h - y) + y(h - y)\frac{2y}{5} \times \frac{y^2}{5} \right] dy$$

$$= 8 \times 10^4 \left[\frac{h^3}{3} - \frac{h^4}{4} + \frac{2}{25}\left(\frac{h^6}{5} - \frac{h^6}{6} \right) \right]$$

$$= 8 \times 10^4 \left(\frac{h^3}{12} + \frac{h^6}{375} \right) = 10^4 \left(\frac{125}{12} + \frac{125}{24} \right)$$

$$= 156.25\text{KN} - \text{m}$$

22.



Total resistance of loop = 6ar

Resistance of left part $R_1 = \left(a + \frac{4x}{\sqrt{3}} \right) r$

Resistance of right part $R_2 = \left(5a - \frac{4x}{\sqrt{3}} \right) r$

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Resistance of middle part $R_3 = \left(a + \frac{2x}{\sqrt{3}}\right)r$

emf induced in middle part $E = BV\left(a + \frac{2x}{\sqrt{3}}\right)$

Current through middle part $I_3 = I_1 + I_2$

$$= BV\frac{\left(a + \left(2x/\sqrt{3}\right)\right)}{R_3 + \left(R_1R_1/R_1 + R_2\right)}$$

Force on rod, $F = BI_3\left(a + \frac{2x}{\sqrt{3}}\right)$

$$= \frac{B^2V\left(a + (2x/\sqrt{3})^2 \times 6ar\right.}{R_1R_2 + R_2R_3 + R_3R_1}$$

$$F = \frac{B^2V}{r} \times \frac{6a\left(a + \frac{2x}{\sqrt{3}}\right)^2}{\left(a + \frac{4x}{\sqrt{3}}\right)\left(5a - \frac{4x}{\sqrt{3}}\right) + \left(5a - \frac{4x}{\sqrt{3}}\right)\left(a + \frac{2x}{\sqrt{3}}\right) + \left(a + \frac{2x}{\sqrt{3}}\right)\left(a + \frac{4x}{\sqrt{3}}\right)}$$

$$= \frac{B^2V}{r} \times \frac{6a\left(a + \frac{2x}{\sqrt{3}}\right)^2}{11a^2 - \frac{16}{3}x^2 + \frac{28}{\sqrt{3}}ax}$$

Power loss $= FV$

$$= \frac{B^2V^2}{r} \times \frac{6a\left(a + \frac{2x}{\sqrt{3}}\right)^2}{11a^2 - \frac{16}{3}x^2 + \frac{28}{\sqrt{3}}ax}$$

$$I_1 - \frac{E - I_3R_3}{R_1} = \frac{E - \frac{ER_3}{R_3 + \frac{R_1R_2}{(R_1 + R_2)}}}{R_1}$$

$$= \frac{E}{R_1} \cdot \frac{\frac{R_1R_2}{(R_1 + R_2)}}{\left(R_3 + \frac{R_1R_2}{(R_1 + R_2)}\right)} = \frac{ER_2}{R_1R_2 + R_2R_3 + R_3R_1}$$

$$I_1 = \frac{BV}{r} \frac{\left(a + \frac{2x}{\sqrt{3}}\right)\left(5a - \frac{4x}{\sqrt{3}}\right)}{11a^2 - \frac{16}{3}x^2 + \frac{28}{\sqrt{3}}ax}$$

$$I_2 = \frac{BV}{r} \frac{\left(a + \frac{2x}{\sqrt{3}}\right)\left(a + \frac{4x}{\sqrt{3}}\right)}{11a^2 - \frac{16}{3}x^2 + \frac{28}{\sqrt{3}}ax}$$

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$$V_{AF} = I_1 ar = BVa\frac{\left(a+\frac{2x}{\sqrt{3}}\right)\left(5a-\frac{4x}{\sqrt{3}}\right)}{\left(11a^2-\frac{16}{3}x^2+\frac{28}{\sqrt{3}}ax\right)}$$

$$V_{CD} = I_2 ar = BVa\frac{\left(a+\frac{2x}{\sqrt{3}}\right)\left(a+\frac{4x}{\sqrt{3}}\right)}{\left(11a^2-\frac{16}{3}x^2+\frac{28}{\sqrt{3}}ax\right)}$$

23. $I_{max} = (\sqrt{I}+\sqrt{4I})^2 = 9I$

$I_{min} = (\sqrt{4I}-\sqrt{I})^2 = I$

$\beta = \frac{\lambda D}{d} = \frac{6000\times10^{-10}\times1}{0.1\times10^{-3}}m = 6mm$

Phase difference at O is $(2n+1)\frac{\pi}{2}$

Path difference at O is $\frac{\lambda}{4}(2n+1)$

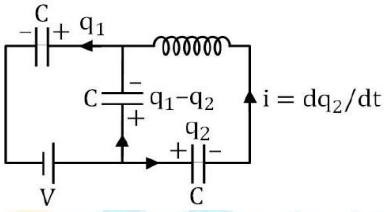
$d\sin30^\circ + (\mu-1)t_A = (\mu-1)t_B = \frac{\lambda}{4}(2n+1)$

$100\times\frac{1}{2}\mu\ m + 0.5(20.4-t_B)\mu m = 0.15(2n+1)\mu m$

$100+20.5-t_B = 0.6n+0.3$

$t_B = 120.1-0.6n$

24.



$\frac{q_1}{C}+\frac{q_1-q_2}{C} = V \Rightarrow 2q_1-q_2 = CV$

$\frac{q_2}{C}+L\frac{di}{dt} = \frac{q_1-q_2}{C}$

$LC\frac{di}{dt} = \frac{CV+q_2}{2}-2q_2 = \frac{CV}{2}-\frac{3}{2}q_2$

$LC\frac{d^2q_2}{dt^2} = -\frac{3}{2}\left(q_2-\frac{CV}{3}\right)$

$\frac{d^2q_2}{dt^2} = -\frac{3}{2LC}\left(q_2-\frac{CV}{3}\right)$

$q_2-\frac{CV}{3} = q_0\sin\left(\sqrt{\frac{3}{2LC}}t+\phi\right)$

$i = q_0\sqrt{\frac{3}{2CL}}\cos\left(\sqrt{\frac{3}{2LC}}t+\phi\right)$

at $t=0, i=0, q_2=0$

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$\Rightarrow \cos\phi = 0 \Rightarrow \phi = \frac{\pi}{2}$ and $-\frac{CV}{3} = q_0$

$\Rightarrow q_2 = \frac{CV}{3}\left(1-\cos\sqrt{\frac{3}{2LC}}t\right)$

Voltage across C_2

$V_2 = \frac{q_1 - q_2}{C} = \frac{CV - q_2}{2C}$

$= \frac{V}{2} - \frac{V}{6}\left(1-\cos\sqrt{\frac{3}{2LC}}t\right)$

$= \frac{V}{3} + \frac{V}{6}\cos\left(\sqrt{\frac{3}{2LC}}\times t\right)$

$V_{2\,min} = \frac{V}{3} - \frac{V}{6} = \frac{V}{6}$

$i_{max} = \left(\frac{CV}{3}\right)\sqrt{\frac{3}{2LC}} = V\sqrt{\frac{C}{6\,L}}$

25. $z = \frac{e}{m} = \frac{8\,V}{r^2\,B^2} = \frac{8\,V}{r^2\mu_0^2 n^2 I^2}$

$\frac{\Delta z}{z} = \frac{1}{20} + 2\times\frac{0.1}{3.4} + 2\times\frac{0.02}{1.92}$

$= \frac{1}{20} + \frac{1}{17} + \frac{1}{48} = 0.1296$

26. $Q_2 = \sigma A T_2^4$

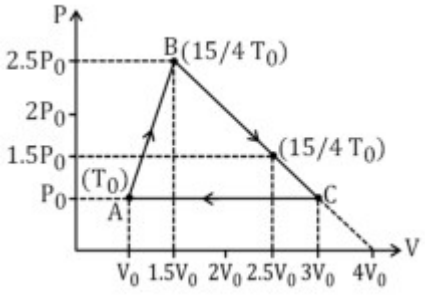
$W = 1\text{ kW}$

$\frac{W}{Q_2} = \frac{Q_1 - Q_2}{Q_2} = \frac{T_1 - T_2}{T_2}$

$1\text{ kW} = \alpha A T_2^4\left(\frac{T_1}{T_2} - 1\right)$

T_2 must be known to solve.

27.



Part BC :

$\frac{dQ}{dV} > 0$

For $V < 5\times\frac{4\,V_0}{8} \Rightarrow V < 5\frac{V_0}{2}$

$W = \frac{1}{2}\times\frac{3}{2}P_0\times 2\,V_0 = \frac{3}{2}P_0\,V_0$

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$$Q_1 = \frac{1}{2}\times\frac{V_0}{2}\times\left(P_0+\frac{5}{2}P_0\right)+n\frac{3R}{2}\left(\frac{15}{4}T_0-T_0\right)+\frac{1}{2}\times V_0\times\left(\frac{5}{2}P_0+\frac{3}{2}P_0\right)+n\frac{3R}{2}\left(\frac{15}{4}T_0-\frac{15}{4}T_0\right)$$

$$= P_0\ V_0\left(\frac{7}{8}+\frac{33}{8}+2+0\right) = 7P_0\ V_0$$

$$\Rightarrow \eta = \frac{W}{Q_1}\times 100\% = \frac{3/2}{7}\times 100\% = \frac{150}{7}\%$$

$$= 21.43\ \%$$

28. At the lowest position, particle is vertically above the centre of disc,

Loss of PE $=$

$$mg\left(\left(2R-\frac{R}{4}\right)-\left(2R+\frac{R}{4}\right)\cos 60^\circ\right)$$

$$= mgR\left(\frac{7}{4}-\frac{9}{8}\right) = \frac{5}{8}mgR$$

$\Rightarrow$ speed of particle at this instant

$$V = \sqrt{\frac{5}{4}gR}$$

Angular velocity of disc at this instant

$$\omega = \frac{V}{R+\frac{R}{4}} = \frac{4\,V}{5R} = \sqrt{\frac{4}{5}\frac{g}{R}}$$

Acceleration of centre of disc,

$$a_0 = \frac{(\omega R)^2}{2R} = \frac{\omega^2 R}{2} = \frac{2\,g}{5}\ \text{upward}$$

Acceleration of particle w.r.t. disc,

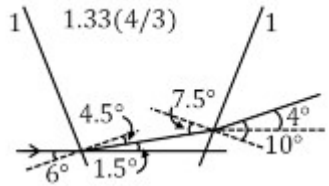
$$a_{p_0} = \frac{\omega^2 R}{4} = \frac{g}{5}\ \text{downward}$$

$\Rightarrow$ Acceleration of particle w.r.t. ground,

$$a_p = \frac{g}{5}\ \text{upward}$$

$$\Rightarrow N = mg + ma_p = \frac{6}{5}mg = 1.2mg$$

29.



Due to vessel, the image is shifted upwards by distance

$$f\times\frac{4\pi}{180} = 50\ \text{cm}\times\frac{\pi}{45} = 3.49\ \text{cm}$$

30. Extension in spring $= y_B - 2y$

Tension in string $= K(y_B - 2y)+\frac{mg}{2}$

Acceleration of $m = \frac{2\,K}{m}(y_B - 2y)$

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$$\Rightarrow \frac{d^2y}{dt^2}+\frac{4\,K}{m}y=\frac{2\,K}{m}\times b\sin(\omega t)$$

$$\ddot{y}+400\,g=200\,b\sin(\omega t)$$

Resonance angular frequency $\omega=\sqrt{400}=20$

31. $\mu=\frac{q}{2\,m}\times I\omega$

$$=\frac{q}{2\,m}\omega\left(\frac{m}{2}\frac{\ell^2}{3}+\frac{m}{2}\frac{\ell^2}{12}+\frac{m}{2}\frac{5\ell^2}{4}\right)$$

$$=\frac{q\omega\ell^2}{2}\left(\frac{1}{6}+\frac{1}{24}+\frac{5}{8}\right)$$

$$=\frac{q\omega\ell^2}{48}(4+1+15)=\frac{5q\omega\ell^2}{12}$$

$$\alpha=\frac{5}{6}=0.83$$

32. $\hat{r}=\frac{3\hat{i}+4\hat{j}}{5}$

$$V_{Sr}=\vec{V}_S\,\hat{r}=30\left(\hat{i}+\hat{j}\right)\times\frac{1}{5}\left(3\hat{i}+4\hat{j}\right)=42$$

$$V_{Or}=\vec{V}_O\,\hat{r}=15\left(\hat{i}+\sqrt{3}\,\hat{j}\right)\times\frac{1}{5}\left(3\hat{i}+4\hat{j}\right)=3\left(3+4\sqrt{3}\right)$$

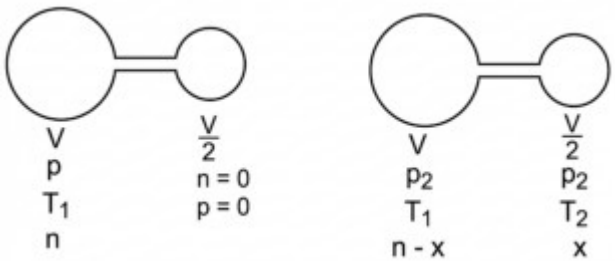
$$f_{app}=1000\text{ Hz}\times\frac{330-3\left(3+4\sqrt{3}\right)}{330-42}$$

$$\frac{f_{app}}{100}=10\times\frac{321-12\sqrt{3}}{288}\text{ Hz}=10.424\text{ Hz}$$

CHEMISTRY

33. 2,4-DNP reacts with carbonyl compounds

34.



$$pV=nRT_1$$

$$p_2\,V=(n-x)RT_1$$

$$p_2\frac{V}{2}=xRT_2$$

$$\frac{pV}{RT_1}=\frac{p_2\,V}{RT_1}+\frac{p_2\,V}{2RT_2}$$

$$\Rightarrow p_2=\frac{2pT_1\,T_2}{T_1(\,T_1+2\,T_2)}$$

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35. for node $\psi^2 = 0$,

$\therefore \quad 2 - Zr = 0$

$r = \frac{2}{Z}\left(He^+, Z = 2\right)$

$r = 1$

36. $NO_3^- \xrightarrow[KOH]{Zn/Al} NH_3 \uparrow$

37. $Na + C + N + S \xrightarrow{\Delta} NaSCN$

$NaSCN + FeCl_3 \rightarrow \left[Fe(SCN)\right]Cl_2 + NaCl$

blood red colour

$[Fe(SCN)]^{+2} + 3C_2O_4^{2-} \rightarrow \left[Fe(C_2O_4)_3\right]^{3-} + SCN^-$

38. Biuret test is given by Proteins, Polypeptides, Heating urea

39.

Process BC $: \frac{P_B}{T_B} = \frac{P_C}{T_C}$

$\Rightarrow \frac{P_B}{500} = \frac{1}{250} \Rightarrow P_B = 2bar$

and $\Delta U_{BC} = n.C_{V_i\,m}\left(T_C - T_B\right)$

$= 2 \times 1.5R \times (250 - 500)$

$= -750R$

Process CD : $\Delta U = w$

$= n \cdot C_{V_i\,m}\left(T_D - T_C\right)$

$= -P_{ext}\left(V_D - V_C\right)$

or, $n \times 1.5R \times \left(T_D - T_C\right)$

$= -P_D\left(\frac{nRT_D}{P_D} - \frac{nRT_C}{P_C}\right)$

$\Rightarrow T_D = 450\ K$

and $\Delta H_{CD} = n.C_{P_i\,m}.\left(T_D - T_C\right)$

$= 2 \times 2.5R \times (450 - 250)$

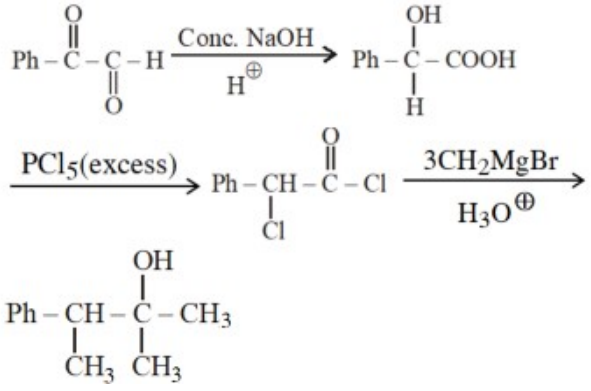
$= 1000R$

40. $H_2(2) = \sigma(1s)^1 \sigma^*(1s)^1$

Bond order $= 0$

So this molecule will not exist and will break down into atoms.

41.



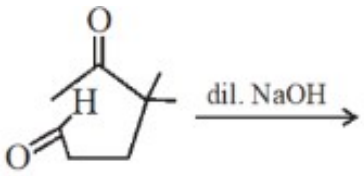
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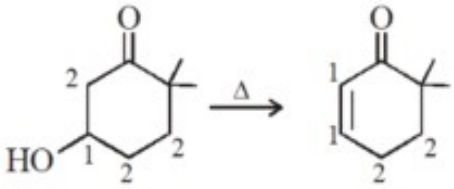
Total moles of CH_3MgBr used is 3 .

Hence total mass= 3×119

$=357$

42.





[A] [B]

Ring H'S in A = 7 Ring H'S in B = 6

Total number of H'S = 7 + 6 = 13

43. (a), (b) and (c) do not give Cannizaro rx^n -

44. Option (ii) and (iv) are correct

$$n=1\Rightarrow t_{100\%}=\frac{1}{K}\cdot\ln\frac{[A_0]}{0}=\text{ infinite}$$

$$n\neq1\Rightarrow t_{100\%}=\frac{[A_0]^{1-n}-(0)^{1-n}}{K(1-n)}=\frac{[A_0]^{1-n}}{K(1-n)}$$

if $n<1$

$n>1\Rightarrow t_{100\%}=$ infinite

45. $A_2\rightleftharpoons 2\ A; K_1=x$ atm

Initial partial pressure 1 atm 0

Equ. Partial pressure $1-(x+z)\ 2x$

$B_2\rightleftharpoons 2\ B; K_2=y$ atm

Initial partial pressure 1 atm 0

Eq. partial pressure $1-(y+z)\ 2y$

$A_2 + B_2\rightleftharpoons K_3=2$

Initial partial	1	1	0
Equ.partialpressure	$1-(x+z)$	$1-(y+z)$	$2z=0.5(1)$

$\therefore x+y=0.75\#(2)$

Now, $K_3=\dfrac{(0.5)^2}{(0.75-x)(0.75-y)}=2$

$\Rightarrow$ $x=0.25$ or 0.50

$y=0.50$ or 0.25

$$\therefore\ \frac{K_2}{K_1}=\frac{\frac{(2y)^2}{1-(y+z)}}{\frac{(2x)^2}{1-(x+z)}}=\frac{(2y)^2\times(0.75-x)}{(2x)^2\times(0.75-y)}=\frac{1}{8}\text{ or }\frac{8}{1}$$

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46. cell reaction $H_{2(g)} + I_{2(s)} \rightleftharpoons 2H^+_{(aq)} + 2I^-_{(aq)}$

$$E_{cell} = E^0_{cell} - \frac{0.0591}{2}\log\frac{\left[H^+\right]^2\left[I^-\right]^2}{P_{H_2}}$$

$$0.7714 = 0.535 - \frac{0.0591}{2}\log\frac{\left(H^+\right)^2(0.1)^2}{1}$$

$pH = 3$

47. (i) $A = 27$ (Cobalt)

(ii) $M(en)_2Cl_3 \rightarrow \left[M(en)_2Cl_2\right]^+ + Cl^-$

$t^6_{2g}, e^0_g\ (B = 6)$

(iii) $M(AA)_2\ b_2$

C = geometrical isomers = 2

D = optical isomers = 2

(iv) 1 mol AgCl ppt. $\Rightarrow 1Cl^-$ outside (E)

(v) $CFSE = -2.4\Delta_o + 2PE$

$A + B + C + D + E + F$

$= 27 + 6 + 2 + 2 + 1 + 2.4 = 40.40$

48. $(5\ mol)$

$P_4 + 3KOH\ (aq.) + 3H_2O \rightarrow PH_3\ (5\ mol) +$

$3KH_2PO_2\ (15\ mol)$

$3H_2PO_2^- + 3H^+ \rightarrow 3H_3PO_2$

$4AgNO_3 + 2H_2O + H_3PO_2\ (15\ mol) \rightarrow 4Ag\downarrow$

$(60\ mol) + 4HNO_3 + H_3PO_4$

Mass of PH_3 :-

Moles of P_4 = moles of PH_3

$$\frac{620}{124} = \frac{x}{34}$$

$$x = \frac{670 \times 34}{124}$$

$x = 170gm$

$3\times$ moles of P_4 = moles of H_3PO_2

$$3 \times \frac{620}{124} = \text{moles of } H_3PO_2$$

$4\times$ moles of H_3PO_2 = moles of Ag

$$\frac{4 \times 3 \times 620}{124} = \frac{\text{mass of Ag}}{108}$$

$$\text{mass of Ag} = \frac{4 \times 3 \times 620 \times 108}{124}$$

mass of Ag $= 6480gm$

Final Answer $= 170gm + 6480gm$

Final Answer $= 6650gm$

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